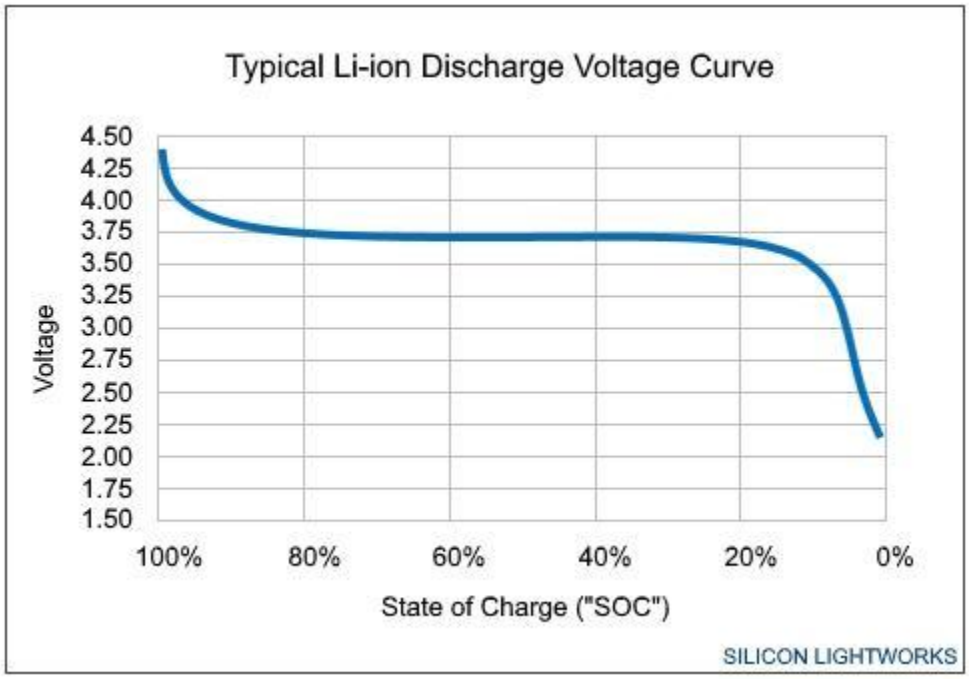
April 4

Brake System Plausibility Device

BSPD: Throttle gives power say >5kW (current sensor{Hall effect} HTFS)
More throttle-> More current ->More power



We don't want the curve to reach the lower plateau region because then it could decrease battery life.

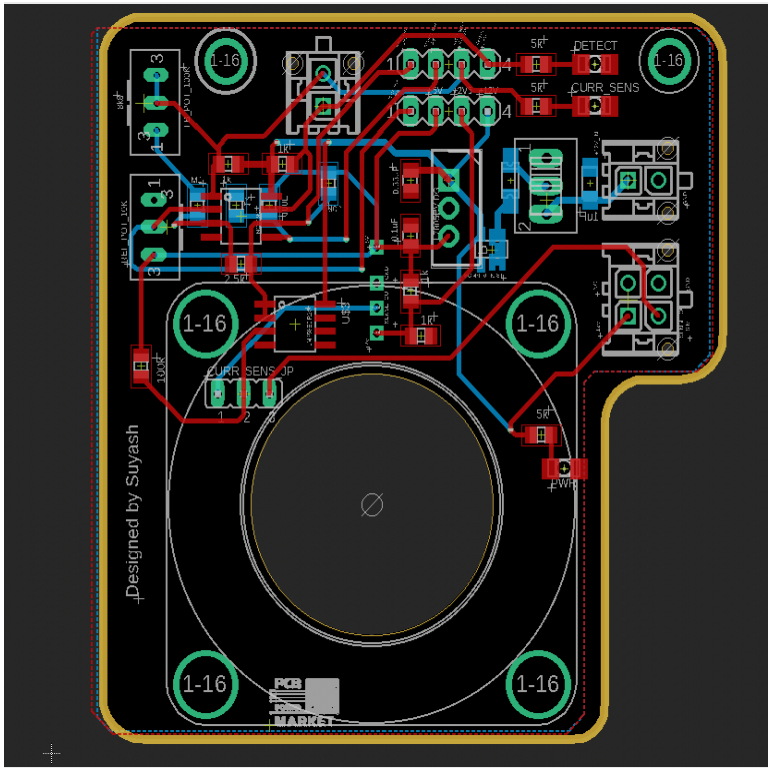
[Search for State of charge curve]

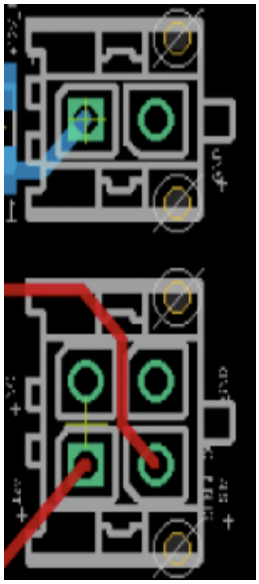
Detect if breaks are also applied (break pressure sensor)

If both yes, then there's an error.

Power block and other blocks are connected by jumper, it is like a mediating path

16 mil routing for everything except power signals is used in our team.





<--Connectors

Jumper caps are used, 2 pin headers can be shorted [2 signals or blocks can be isolated]

Dimension layer is the boundary, only the stuff inside it is printed on the actual PCB. [Yellow boundary in the above photo]

No acute angles while connected

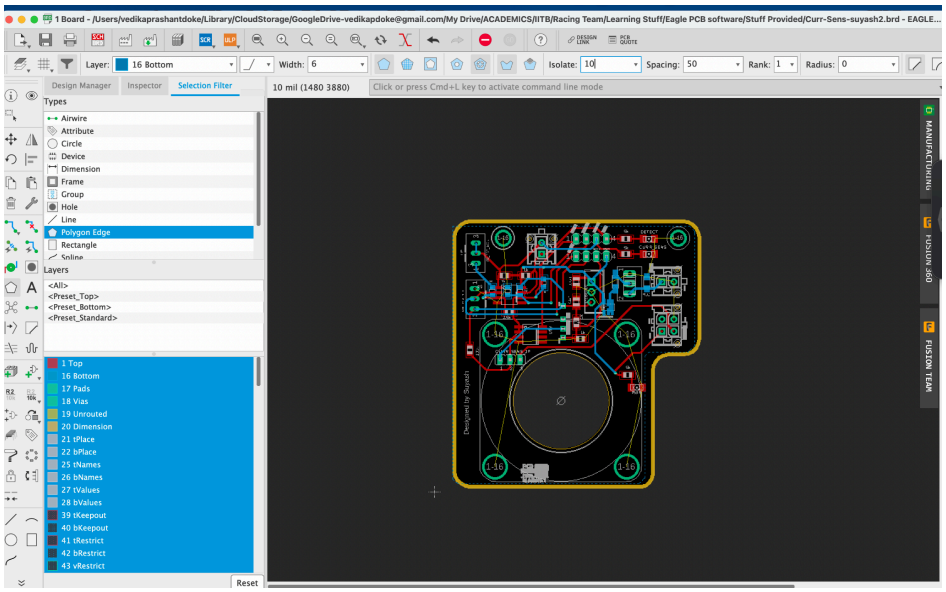
Theoretical reason: CrossTop, the less space between the two signals will cause impedance.

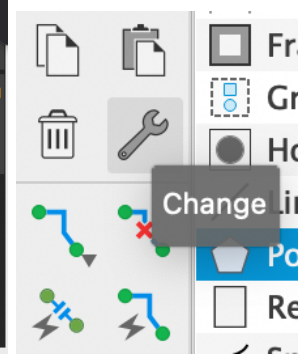
Manufacturing reason: NO 90 degree or acute angle, they aren't printed that accurately While manufacturing.

Polygon properties-> Parameter called Isolate-> has to be 10

For ground plane (big area)-> Heat will dissipate easily

To fix this [Change option-> Thermals-> ON] extra part is added when thermals is set ON, so we get some extra area which won't be connected to the large ground area and Hea will basically be localised





Thumb rule: width of trace=x, 3x should be distance between next trace. [But nobody follows this]

Keep the components far enough so that you are able to solder stuff.
Make 'em blockwise. Make it easy to understand.

Push the name/value to tplace layer. (Instead of actually typing out the component name and values again)

Use a mirror tool to change the layer.

Place via tool and then connect to ground. (connects two layers)
(google what Island is)
Through holes should all be in the top layer.

DRC Check:

Tools-> DRC

BEFORE April 7th Upload explanation of given schematic on github
BEFORE April 13th replicate given pcb board.
(Add curves at the edges)